

Coupon Collector Problem

How long does it take to collect every type?

MATH 394 Probability I Mini-Project



The question

A simple question with a surprisingly logarithmic answer

Imagine cereal boxes. Each box contains one random coupon type. There are n types, and each type is equally likely.

Goal: keep opening boxes until every coupon type has appeared at least once.

Type 1

Type 2

Type 3

Type 4

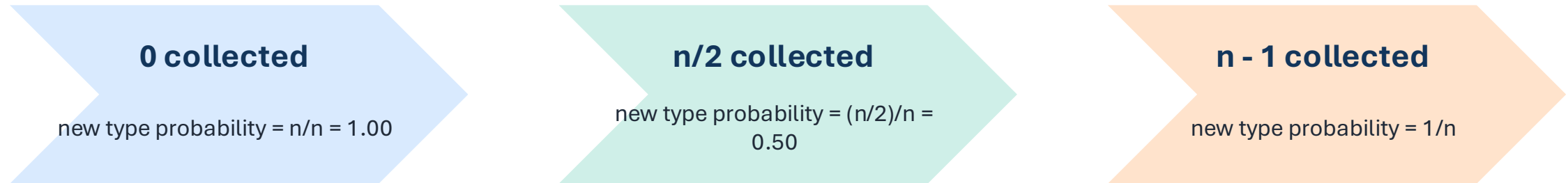
Type 5

Type 6

Random draw with replacement

Why duplicates matter

The probability of finding a new type changes as the collection grows



Early draws are almost always useful. Near the end, most draws are duplicates, so the last few coupon types dominate the waiting time.

Break the waiting time into stages

Use geometric random variables and linearity of expectation

Let T be the total number of draws needed.

Let X_k be the number of additional draws needed to get the k -th new coupon type, after $k - 1$ types have already been collected.

Probability next draw is new at stage k :

$$p_k = (n - k + 1) / n$$

$$X_k \sim \text{Geometric}(p_k), \text{ so } E[X_k] = 1 / p_k = n / (n - k + 1)$$

Main result

The expected collection time is n times a harmonic sum

$$E[T] = E[X_1 + X_2 + \dots + X_n] = n(1 + 1/2 + 1/3 + \dots + 1/n) = nH_n$$

For large n , $H_n \approx \ln(n) + 0.577$, so $E[T] \approx n \ln(n) + 0.577n$.

Key idea: collecting all n types grows faster than n . The extra $\log(n)$ factor comes from repeatedly waiting for rare missing types near the end.

Worked example: 5 coupon types

How many boxes do we expect to open?

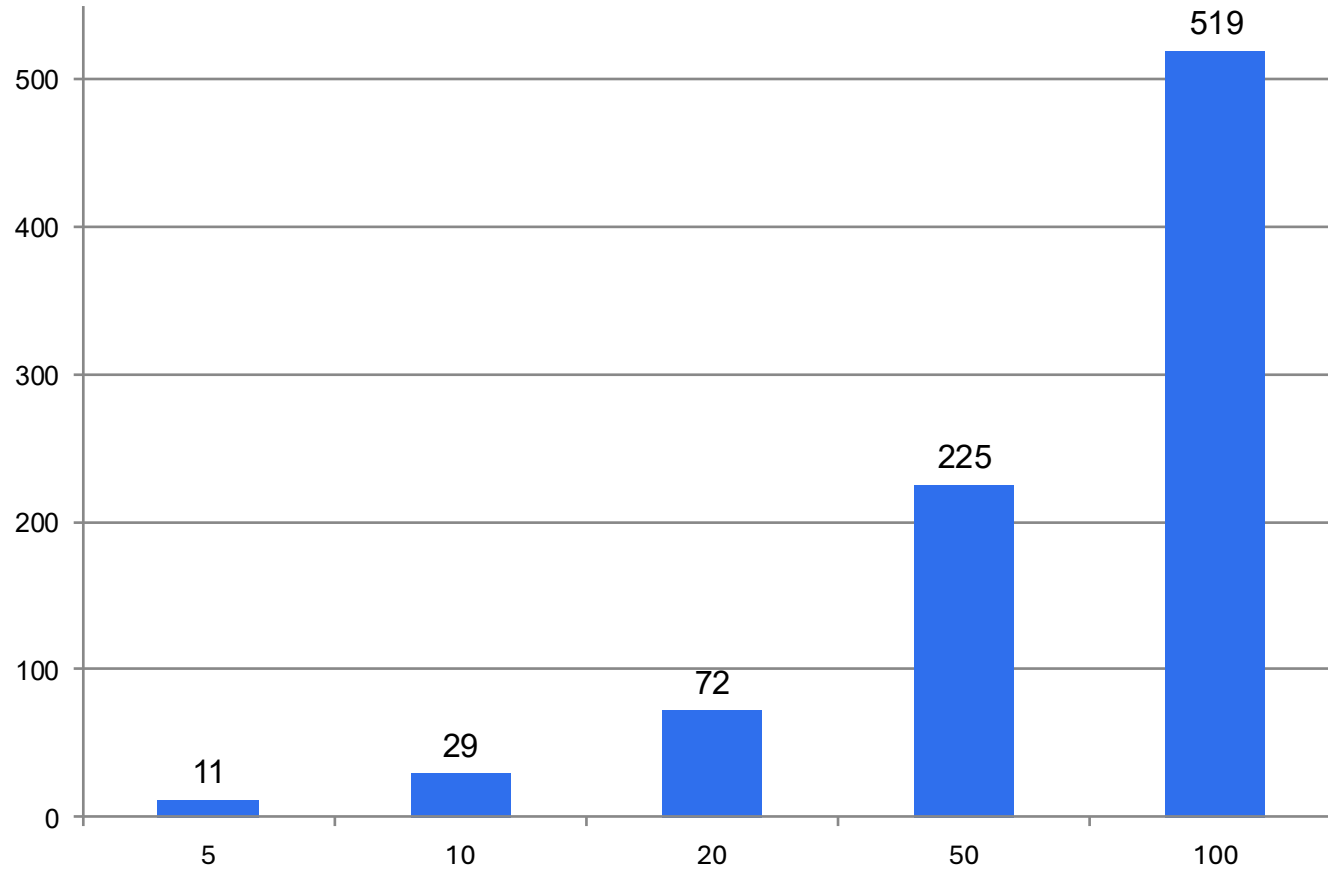
Stage	New-type probability	Expected wait
1st	5/5	1
2nd	4/5	$5/4 = 1.25$
3rd	3/5	$5/3 \approx 1.67$
4th	2/5	$5/2 = 2.5$
5th	1/5	5

$$\begin{aligned} E[T] &= 5(1 + 1/2 + 1/3 + 1/4 + 1/5) \\ &= 5 \times 2.2833 \\ &\approx 11.42 \text{ boxes} \end{aligned}$$

Even with only 5 types, we need more than twice as many boxes as types on average.

Growth pattern

Expected draws for different numbers of coupon types



**The average time is about $n \ln(n)$, not just n .
Doubling the number of types more than doubles the expected waiting time.**

Diminishing returns

A probability question too

What is the chance we have completed the set by draw m ?

For $n = 5$ types, exact completion by m draws can be counted using inclusion-exclusion:

$$P(\text{all 5 seen by } m) = 1 - C(5,1)(4/5)^m + C(5,2)(3/5)^m - C(5,3)(2/5)^m + C(5,4)(1/5)^m$$

m	5	8	10	12	15
$P(\text{complete})$	0.038	0.322	0.522	0.693	0.829

Expectation is not a guarantee. Around 11-12 boxes is the average, but there is still a real chance the set is unfinished.

Applications

The same structure appears whenever we wait to “see every type”

Testing

How many random tests until every category of bug or case appears?

Sampling

How many observations until every species or category has been seen?

Computing

How long until every cache line, page, or randomized item has been touched?

The exact “coupon” can be cereal cards, bugs, species, page requests, or any categories sampled randomly with replacement.

Variants and limitations

The standard result depends on equal probabilities

Standard model assumptions:

- Every draw is independent
- Each coupon type is equally likely
- We only need one copy of each type

Common variants:

- Unequal probabilities: rare coupons matter even more
- Need multiple copies of each type
- Stop after collecting k out of n types

Takeaway: the “last few missing types” are usually the bottleneck.

Conclusion

One clear idea to remember

The Coupon Collector Problem asks how long it takes to randomly draw until every one of n types has appeared.

$$\text{Expected time} = nH_n \approx n \ln(n) + 0.577n$$

The proof works by splitting the total time into geometric waiting stages and adding their expected values. The slow part is finding the last few missing types.

Sources

- MATH 394 Mini-Project instructions, University of Washington, Department of Mathematics.
- Brilliant Math & Science Wiki, “Coupon Collector Problem.”
- Tufts University, “The Coupon Collector Problem,” COMP 250 class notes.
- Wikipedia, “Coupon collector’s problem,” accessed for formula checks and extensions.
- Ferrante, M. and Saltalamacchia, M., “The Coupon Collector’s Problem,” MATerials MATemàtics, 2014.